

Assignment 01

Bayesian models & data analysis

2026-01-18

Part 1: Sets

1.1 You are given two sets: $A = \{x|x \in \mathbb{R} \text{ and } -1 \leq x \leq 2\}$ and $B = \{x|x \in \mathbb{R} \text{ and } 1 \leq x \leq 4\}$. Find the following

- (a) $A \cup B$
- (b) $A \cap B$
- (c) $A \setminus B$
- (d) $\bar{A} \cap B$, where $\bar{A}$ represents the complement of A such that $\bar{A} = \{x|x \in \mathbb{R} \text{ and } x \notin A\}$

Part 2: Probability

2.1 In an experiment, two coins a and b are tossed simultaneously. You are given that the probability of getting heads on the coin a is 0.6 and the probability of getting heads on the coin b is 0.4.

You are also given that the outcomes of the coin a and b are independent of each other, i.e., the outcome of coin a does not affect the outcome of coin b and vice versa.

- (a) What is the sample space of the experiment? (What are all the possible outcomes that can occur when the coins a and b are tossed?)
- (b) Consider two events E_1 and E_2 in the event space of the experiment.
 - E_1 : heads appear on either coin a or coin b or both
 - E_2 : heads appear only on coin a
 - i. What is the probability of occurrence of event E_1 ?
 - ii. What is the probability of occurrence of event E_2 ?
- (c) Calculate the probability of occurrence of each elementary event (each possible outcome) in the sample space of the experiment.

(Hint: When two outcomes, say H_a and H_b , are independent of each other, the (joint) probability of co-occurrence of these outcomes $P(H_a H_b)$ will be equal to $P(H_a) \cdot P(H_b)$. For example, in the above experiment, you can calculate the probability of obtaining heads on both the coins as $P(H_a H_b) = P(H_a)P(H_b) = 0.6 \times 0.4$. Similarly, the probability of getting heads on coin a and tails on coin b will be $P(H_a T_b) = 0.6 \times (1 - 0.4)$)

- 2.2 A shooter tries to hit a target 5 times. Suppose the shooter misses the target by following margins (in centimetres) in 5 trials:

5.76, 4.56, 2.33, 6.32, 5.4

What is the sample space of the experiment if you want to analyze the accuracy of the shooter in terms of distance from the target?

Consider a random variable X . The random variable X maps the above sample space to a discrete sample space Ω_x such that the target missed by a margin of greater than 5 centimetres is a “failure” otherwise it is a “success”. And, an outcome in the sample space Ω_x represents the number of successes out of 5 trials.

- (a) Describe the random variable X as a function of ω , where ω is member of the sample space Ω and it is the set of all five margins (in cm) by which the shooter misses the target in a trial.
- (b) Define a probability distribution for the random variable X .
(Hint: The probability distribution of a random variable X is the probability-assigner function that assigns probability values to all possible values of the random variable X . In other words, it is the function that assigns probabilities to all possible outcomes in the sample space Ω_x .)

Part 3: Discrete random variables

- 3.1 In a visual word recognition experiment, a participant has to recognize whether the word shown on screen is a meaningful word (e.g., “book”) or a non-word (e.g., “bktr”). The participant is asked to answer “yes” if the word is meaningful, and “no” if it is a non-word. Suppose a participant is shown 50 words, what is the probability that the participant will correctly recognize 45 out of 50 words? You are given that the probability of correctly recognizing a word is 0.9.

(Hint: Use the binomial distribution, $f(k, n, p) = \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}$. In this question, you are given $k = 45$, $n = 50$, $p = 0.9$.)

- 3.2 Suppose in a small city, 10 road accidents happen on average in a single day. The probability of k number of road accidents in a day is given by the probability mass function $f(k, \lambda) = \frac{\lambda^k e^{-\lambda}}{k!}$ where $\lambda = 10$. What is the probability of zero road accidents in a day?

Part 4: The rate of COVID infections

Suppose some researchers are recording the infection rate of COVID in Europe. They assume that the number of covid infections in a day comes from a true, underlying generative process. They of course do not know what is that generative process.

A researcher claims that she knows about this generative process. She describes the generative process as follows:

- Suppose the number of infections in a day is k .
- The probability that k infections would occur in a day is given by the following function:

$$P(X = k) = f(k) = \frac{10^k e^{-10}}{k!}$$

where f is the probability-assigner function, called the **probability mass function**.

For example, the probability that just 3 infections would happen in a day can be calculated as $\frac{10^3 e^{-10}}{3!} = \frac{10^3 e^{-10}}{3 \times 2 \times 1} = 0.0075$

You can write this probability-assigner function in R.

```
# The following function implements the probability mass function
# of the generative process claimed by the researcher
```

```
PMF_generative_process_researcher_claim <- function(k){
  probability_value <- ((10^k)*(exp(-10)))/factorial(k)
  return(probability_value)
}
```

- 4.1 What is the probability that 100 infections occur in a day?
- 4.2 What is the probability that the number of infections would be somewhere between 5 and 15?
- 4.3 What is the sample space of covid infection rate (observation) i.e., what are possible values that can be reported as "number of infections in a day"?
- 4.4 Draw the graph of the probability mass function $f(k)$.
(Hint: Choose a range of possible values for k which is the number of infections in a day; compute the probabilities for each value of k using the above R function; and then, plot the graph with probabilities on y -axis and k on x -axis.)
- 4.5 Suppose the actual infection count reported in the last ten days is 18, 20, 23, 25, 22, 30, 11, 10, 23, 12. Do you think the researcher's claim can be correct?
- 4.6 The following R function generates the number of infections in a day based on the generative process claimed by the researcher. (Note that this function is not a probability assigner function; it generates the outcomes, not the probabilities.)

```
Generative_process_researcher_claim <- function(n=1){
  infection_rate <- rpois(n,10)
  return(infection_rate)
}
```

The infection-rate data generated by the above function can be called the researcher's prediction.

- 4.6.1 Generate infection-rate predictions for next 5 days based on the researcher's claim.
(Hint: Generate 5 outcomes from the R function using *Generative_process_researcher_claim*(*n* = 5).)
- 4.6.2 Again generate infection-rate predictions for next 5 days based on the researcher's claim. Are the first 5 predictions same as the next 5 predictions?